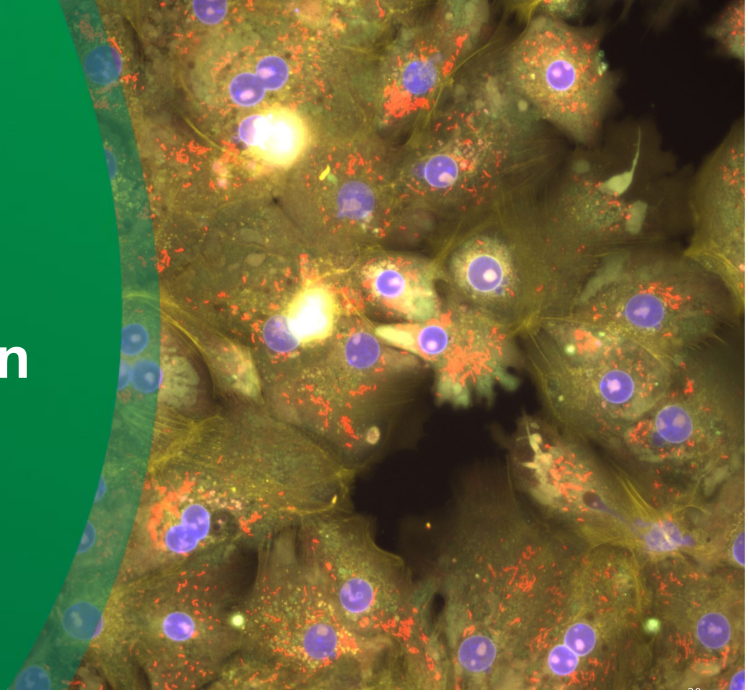
Group project : Bioimaging

Investigating chemical mode-of-action with cell profiling and machine learning



Data science for life scientists 2026

Guillaume Ollitrault

15-19 June 2026

EMBL-EBI

Context

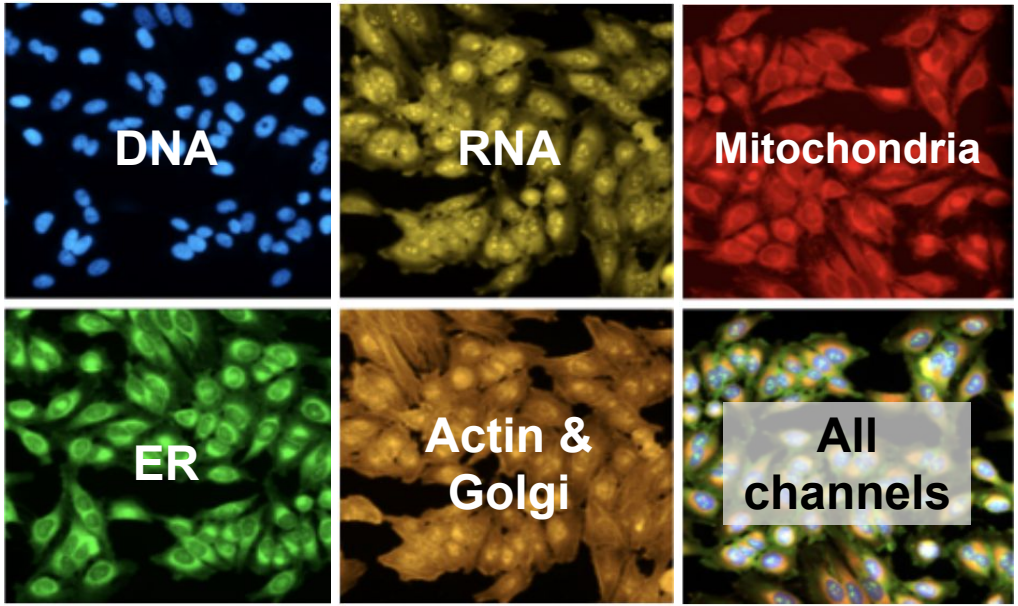
Project:

Work with Cell Painting images of compound-treated primary human hepatocytes to develop machine learning models that predict toxicity modes of action from morphological features.

Learning outcomes:

- Understand the principles of Cell Painting assays
- Visualise, Process and analyse morphological profiles
- Develop and evaluate simple machine learning and deep learning models mode of action prediction
- Interpret bioimaging datasets

The Cell Painting assay is optimized high-content imaging



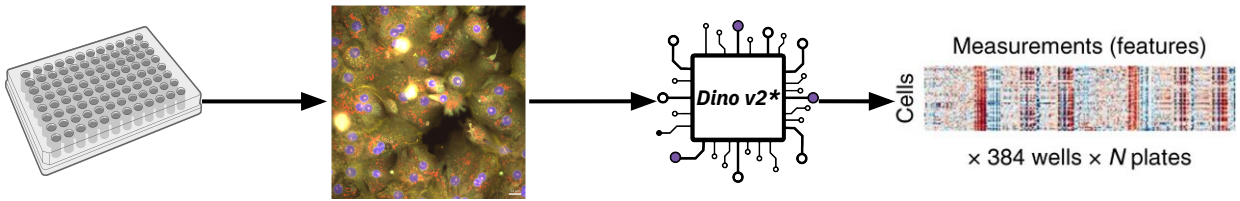
- 6 dyes optimized for:
  - Cost
  - Information content
- Inexpensive:
  - >1000X less than other single-cell omics
  - >15X less than bulk omics
- Automated quantitative analysis
  - Millions-billions of cells

We screened 1085 compounds in primary human hepatocytes

1, 085 reference compounds

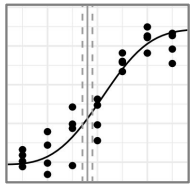
8 Concentrations
44 H exposure
Primary Human Hepatocytes cells

Cell Painting assay

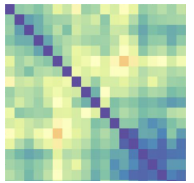


Multiwell plates | Microscopy image | Image analysis (Vision transformer) | Morphological profiles

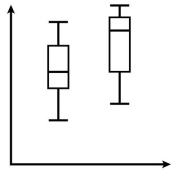
Downstream analysis



Dose-response analysis



Mode-of-action analysis

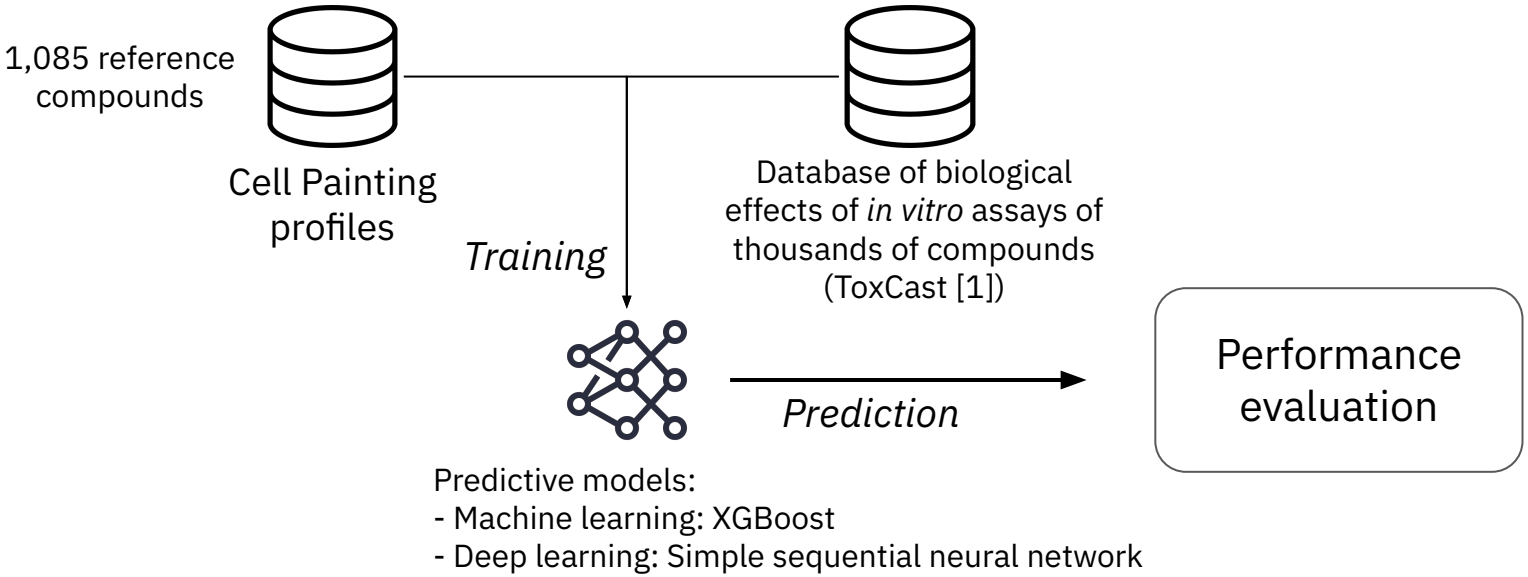


Classification prediction

***Dino v2** (*Oquab et al., 2023*): Vision transformer pretrained on 142 million images yielding high-dimensional image-level feature representations

Ewald et al., 2026. Cell Systems 101566

Mode-of-action prediction



[1] U.S. EPA. 2015. ToxCast & Tox21

Project planning

Can we use Cell Painting morphology profiles to predict chemical cytotoxicity or mode of action?

Course materials:

https://github.com/ewald-lab/2026_DataScienceForLifeScientists_Training

D1: Brainstorming/Propose an analysis workflow to answer this question
D2-4: Go through the prepared notebooks to learn how to process Cell Painting images and develop machine learning predictive models
D5: Prepare a short talk about the project


Thank You